Name : Hamid Reza Ghahramani Date:10 Oct 2026

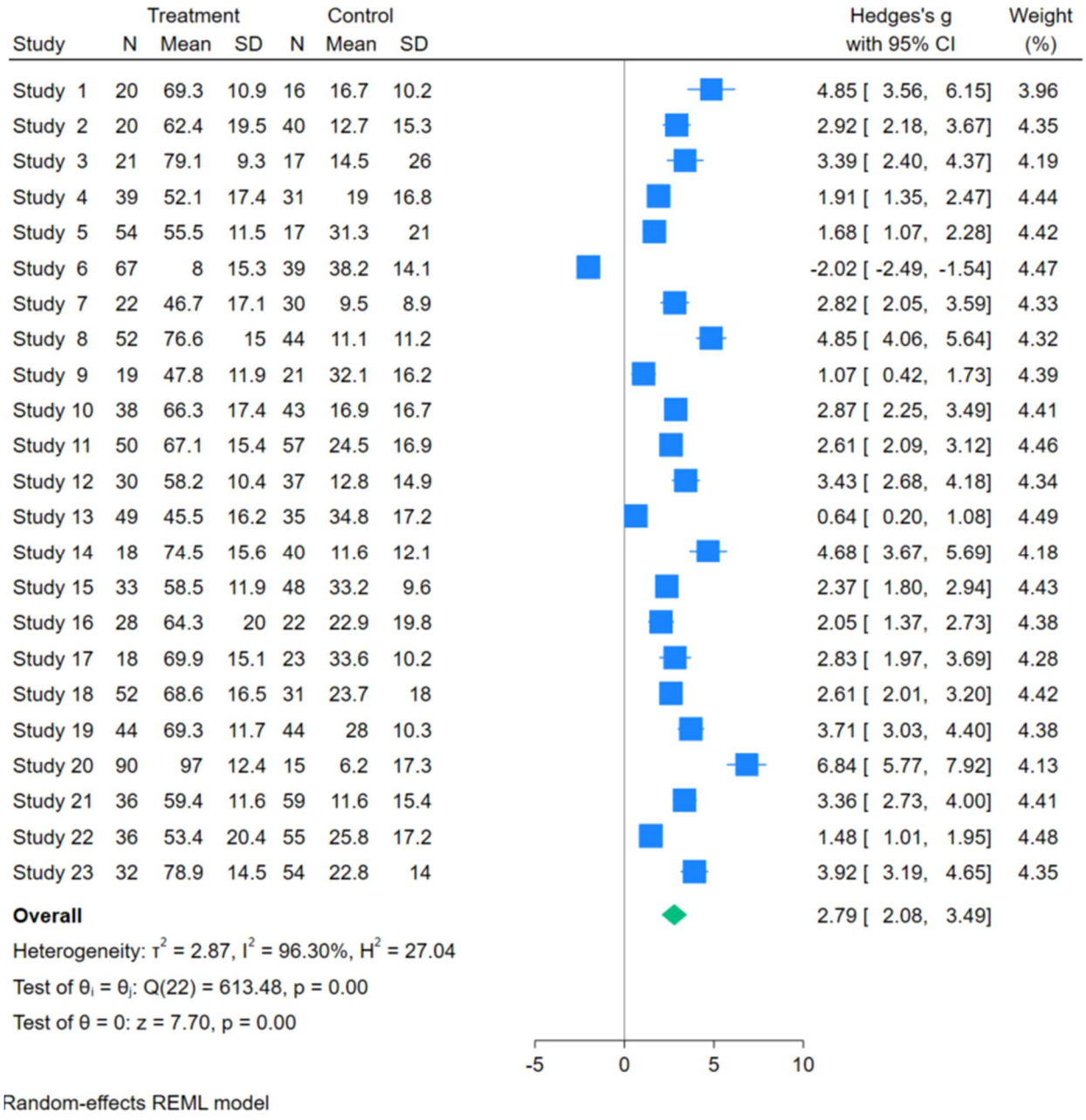
Based on the dataset **"Deep Brain Stimulation Targets in Epilepsy: Systematic Review and Meta-analysis of the Anterior Thalamic Nuclei,"** complete the following tasks:

Task A

1. Identify the outlier(s).
2. Draw and interpret the Galbraith plot.
3. Perform a leave-one-out analysis and interpret the results.
4. Draw and interpret the forest plot both with and without the outlier(s).

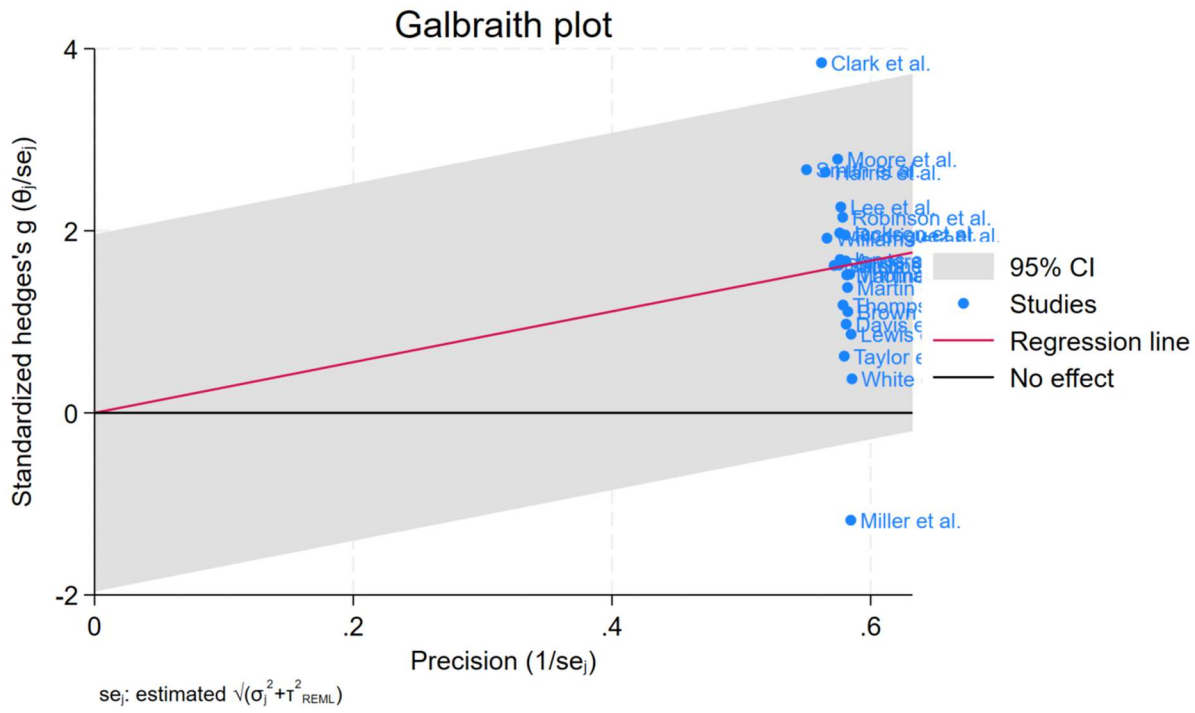
1. Based on the initial visual inspection of the meta-analysis results, Study 6 appears to be a potential outlier. However, this finding requires further investigation using additional statistical methods to confirm whether it is an outlier and to assess its influence on the overall results.

Name : Hamid Reza Ghahramani Date:10 Oct 2026

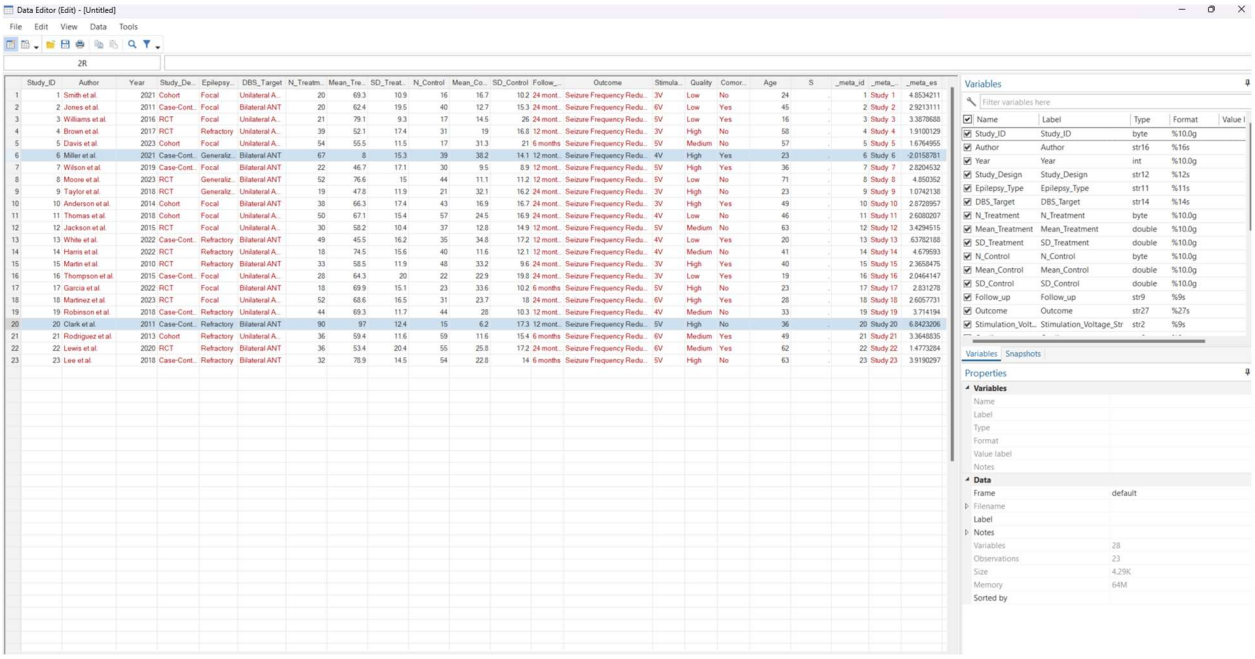


2. After resetting the scale, we can observe that two studies, Miller and Kolar, fall outside the confidence interval. These studies may be potential outliers, so we will investigate them further using additional statistical methods and examine their influence on the forest plot.

Name : Hamid Reza Ghahramani Date:10 Oct 2026



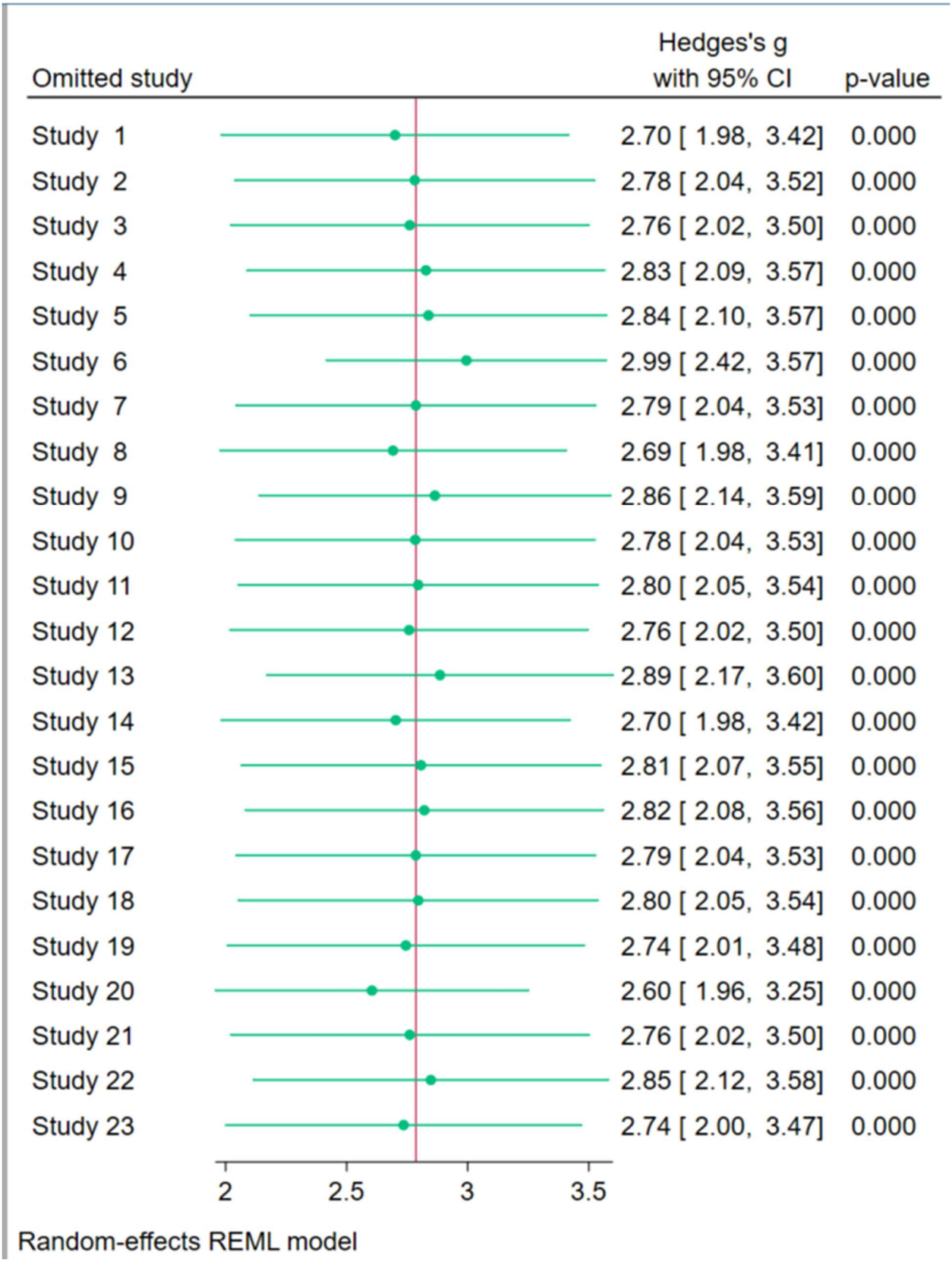
3. Next, we will create a separate section and label the studies that are potential outliers and may need to be excluded from the analysis.



Name : Hamid Reza Ghahramani Date:10 Oct 2026

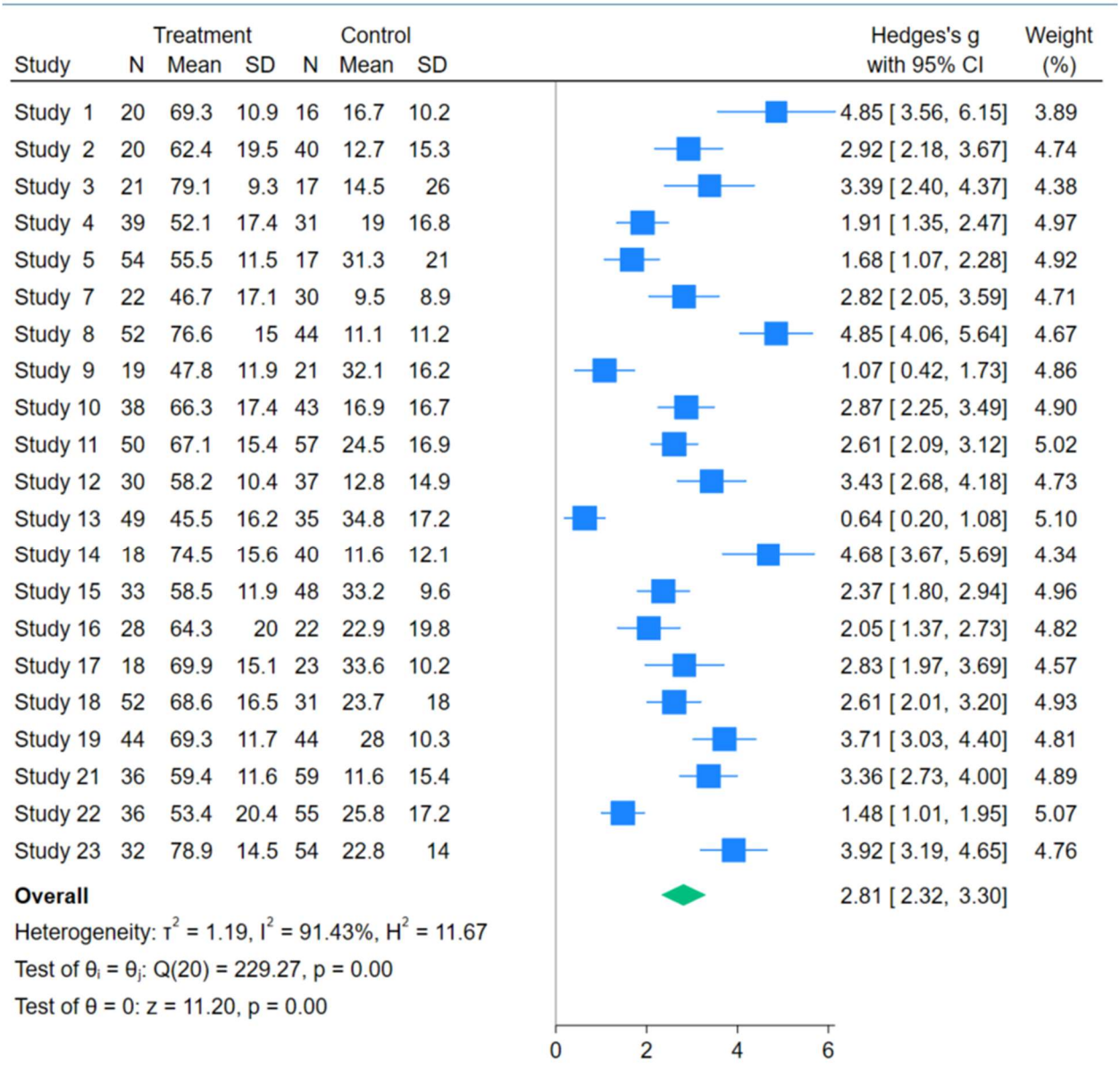
4. During our assessment, we noticed that Study 6, by Miller, had a relatively large influence on the forest plot. We also identified this study as a potential outlier in the Galbraith plot. Therefore, we decided to investigate its influence further by performing a leave-one-out sensitivity analysis and comparing the results with and without this study. This will help us determine whether excluding the study changes the pooled effect size, confidence interval, and heterogeneity. However, we will not exclude the study solely to achieve statistical significance or narrower confidence intervals.

Name : Hamid Reza Ghahramani Date:10 Oct 2026



Name : Hamid Reza Ghahramani Date:10 Oct 2026

5. As shown in the forest plot, after removing the two potential outliers, the overall effect size increased. This suggests that these studies had an influence on the pooled effect estimate.



Task B

Draw the forest plot and, using **each of the three methods learned**, separately assess whether **heterogeneity is present or absent**, and interpret the findings.

(Screenshots should be taken throughout the process)

Name : Hamid Reza Ghahramani Date:10 Oct 2026

Cochran's Q test indicated statistically significant heterogeneity among the 23 studies (Q = 613.48, df = 22, $p < 0.001$). The I^2 value was 96.30%, indicating considerable heterogeneity among the study effect sizes.

```
  Effect-size label: Hedges's g
        Effect size: _meta_es
          Std. err.: _meta_se

Meta-analysis summary                     Number of studies =     23
Random-effects model                      Heterogeneity:
Method: REML                                          tau2 =  2.8663
                                                   I2 (%) =   96.30
                                                       H2 =   27.04
```

Study	Hedges's g	[95% conf.	interval]	% weight
Study 1	4.853	3.561	6.146	3.96
Study 2	2.921	2.177	3.666	4.35
Study 3	3.388	2.402	4.374	4.19
Study 4	1.910	1.346	2.474	4.44
Study 5	1.676	1.071	2.282	4.42
Study 6	-2.016	-2.493	-1.539	4.47
Study 7	2.820	2.054	3.587	4.33
Study 8	4.850	4.057	5.644	4.32
Study 9	1.074	0.422	1.726	4.39
Study 10	2.873	2.254	3.491	4.41
Study 11	2.608	2.094	3.122	4.46
Study 12	3.429	2.679	4.180	4.34
Study 13	0.638	0.197	1.078	4.49
Study 14	4.680	3.666	5.693	4.18
Study 15	2.366	1.795	2.936	4.43
Study 16	2.046	1.366	2.727	4.38
Study 17	2.831	1.970	3.692	4.28
Study 18	2.606	2.013	3.198	4.42
Study 19	3.714	3.027	4.402	4.38
Study 20	6.842	5.770	7.915	4.13
Study 21	3.365	2.734	3.996	4.41
Study 22	1.477	1.009	1.946	4.48
Study 23	3.919	3.190	4.648	4.35
theta	2.786	2.077	3.495	

```
Test of theta = 0: z = 7.70                  Prob > |z| = 0.0000
Test of homogeneity: Q = chi2(22) = 613.48     Prob > Q = 0.0000

.
```

Name : Hamid Reza Ghahramani Date:10 Oct 2026

TASK C

Explore heterogeneity in 3 ways that you have learned for this forest plots.

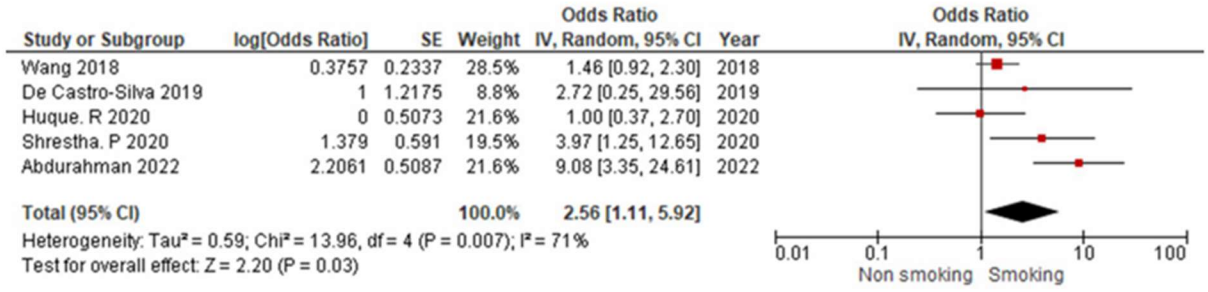


Fig. 2 Forest Plot of the Association between Smoking and Depression among TB patients

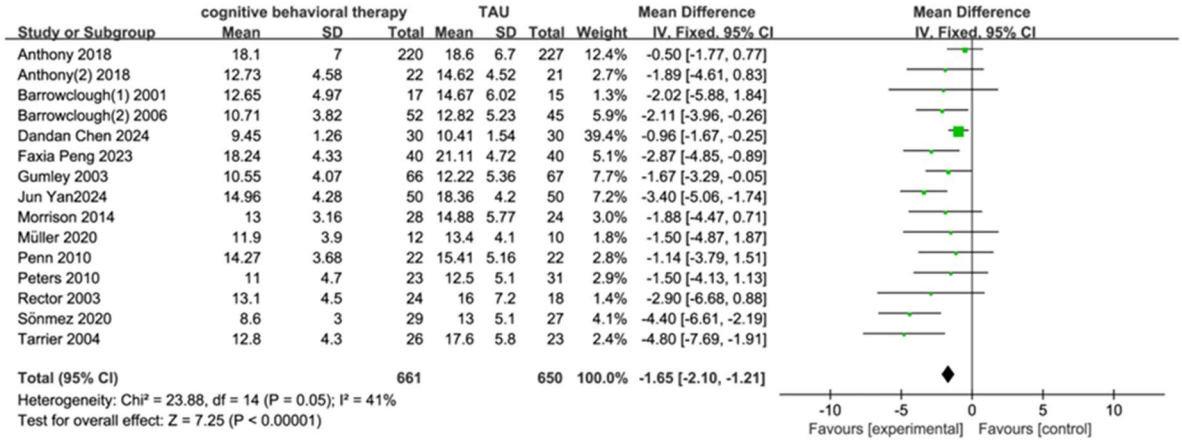


Fig 4. Forest plot of the effect of CBT on negative symptoms of schizophrenia.

The results suggest that Figure 2 shows greater heterogeneity, as indicated by the higher I^2 value and the greater variability in effect sizes across studies. The forest plot also shows considerable variation in the widths of the confidence intervals, with some studies having much wider intervals than others. However, the p-value indicates the statistical significance of heterogeneity rather than its magnitude. Therefore, I^2 and the visual assessment of the forest plot are more useful for comparing the extent of heterogeneity between the two figures.